Name: ______________________

STRAIGHT-LINE GRAPHS, (PARALLEL AND PERPENDICULAR LINES)

Date: ______________

LO: I can identify and find equations of parallel and perpendicular lines.

Parallel and perpendicular lines

Parallel lines have the same gradient ($m_1 = m_2$).
Perpendicular lines have gradients that multiply to give -1.

Key rule: If one line has gradient m, a perpendicular line has gradient $-\frac{1}{m}$ (the negative reciprocal).

Quick examples

●	y = 2x + 1 and y = 2x - 3 are parallel	same gradient m = 2
●	y = 3x and y = $-\frac{1}{3}$x are perpendicular	$3 \times -\frac{1}{3} = -1$
●	Negative reciprocal of 4 is $-\frac{1}{4}$	flip and negate
●	y = 2x + 1 and y = -2x + 3 are parallel	gradients are 2 and -2, not equal

Key idea

Parallel: same gradient. Perpendicular: gradients multiply to -1.

$$m_1 \times m_2 = -1 \qquad m_2 = -\frac{1}{m_1}$$

Negative reciprocal for perpendicular lines

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – identifying parallel lines

Are y = 3x + 5 and y = 3x - 2 parallel?

Gradient of first line = 3

Gradient of second line = 3

Same gradient, so they are parallel

Yes, parallel

Parallel lines have equal gradients but different y-intercepts.

Example 2 – perpendicular gradient

Find the gradient perpendicular to y = 4x + 1.

Original gradient m = 4

Negative reciprocal: $-\frac{1}{4}$

Perpendicular gradient = $-\frac{1}{4}$

$\boxed{m = -\frac{1}{4}}$

Flip the fraction and change the sign.

Example 3 – equation of parallel line

Find the line parallel to y = 2x + 3 through (1, 5).

Parallel means same gradient: m = 2

y = 2x + c, substitute (1, 5): 5 = 2(1) + c

c = 3, equation: y = 2x + 3

$\boxed{y = 2x + 3}$

Use the same gradient and find c from the given point.

Example 4 – equation of perpendicular line

Find the line perpendicular to y = 3x - 1 through (6, 2).

Original gradient = 3, perpendicular = $-\frac{1}{3}$

y = $-\frac{1}{3}$x + c, substitute (6, 2): 2 = -2 + c

c = 4, equation: y = $-\frac{1}{3}$x + 4

$\boxed{y = -\frac{1}{3}x + 4}$

Use the negative reciprocal gradient and substitute the point.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying parallel lines

State whether each pair is parallel, perpendicular, or neither.

a) $y = 5x + 1$ and $y = 5x - 4$

b) $y = 2x$ and $y = -\frac{1}{2}x + 3$

c) $y = 3x + 2$ and $y = -3x + 2$

d) $y = x + 7$ and $y = -x + 1$

2. Negative reciprocals

Find the perpendicular gradient.

a) $m = 2$

b) $m = -5$

c) $m = \frac{3}{4}$

d) $m = -\frac{1}{3}$

3. Equations of parallel lines

Find the equation of the parallel line through the given point.

a) Parallel to $y = 4x + 1$ through (2, 11)

b) Parallel to $y = -x + 5$ through (3, 1)

c) Parallel to $y = \frac{1}{2}x - 3$ through (4, 0)

d) Parallel to $y = 3x$ through (1, 7)

4. Equations of perpendicular lines

Find the equation of the perpendicular line through the given point.

a) Perpendicular to $y = 2x + 5$ through (4, 1)

b) Perpendicular to $y = -3x + 2$ through (3, 5)

c) Perpendicular to $y = \frac{1}{4}x$ through (8, 0)

d) Perpendicular to $y = -x + 6$ through (1, 3)

5. Mixed problems

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Parallel lines have the same gradient ☐	b) The perpendicular gradient of 3 is $\frac{1}{3}$ ☐	c) $m1 \times m2 = -1$ for perpendicular lines ☐	d) $y = 2x + 1$ and $y = -2x + 1$ are perpendicular ☐	e) The negative reciprocal of $-\frac{2}{5}$ is $\frac{5}{2}$ ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Line L has equation $\mathbf{y = 2x - 3}$. Point A is **(4, 5)**. Find the equation of the line through A that is perpendicular to L. Find the point where this perpendicular line meets L.

Expression: ____________________________

Simplified: ____________________________

b) Triangle PQR has vertices **P(0, 6)**, **Q(4, 0)**, and **R(8, 4)**. Show that angle PQR is a right angle by finding the gradients of PQ and QR.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!